

Anonymous Review Draft

} \\ \\ Independent Researcher } Data-driven system identification of complex dynamical systems has advanced rapidly with deep neural surrogates and sparse regression. However, selecting parsimonious models from trajectory data remains challenging because standard validation loss is often insensitive to over-parameterization and noise-fitting shortcuts. In this work, we propose a representation-aware symbolic identification framework that audits internal representation geometry as a selection signal. By integrating Centered Kernel Alignment (CKA), Projection Weighted Canonical Correlation Analysis (PWCCA), and a numerically stable formulation of the Effective Volume of the Covariance Ellipsoid (EV3), our automated pipeline monitors representation stability and geometric complexity across search cycles. We evaluate our framework on chaotic systems (Lorenz, Rössler, pendulum) and physiological signals. Our results show that our pipeline successfully recovers exact numerical coefficients within predefined libraries under dense noise of SNR = 20 dB (achieving 92% recovery on Lorenz, outperforming pure SINDy at 45%). Furthermore, representation auditing successfully identifies concept drift and representation collapse in recurrent models (final layer CKA collapsing to 0.000) under domain shift, while continuous-time systems maintain high stability (CKA of 1.0000). A core limitation of this work is that the search space is limited to predefined libraries, and the language model acts as a heuristic selector rather than an open-ended generator. All results are evaluated on a single-seed basis. representation auditing, symbolic regression, dynamical systems, neuro-symbolic, EV3, system identification. Discovering parsimonious differential equations from trajectory observatio